

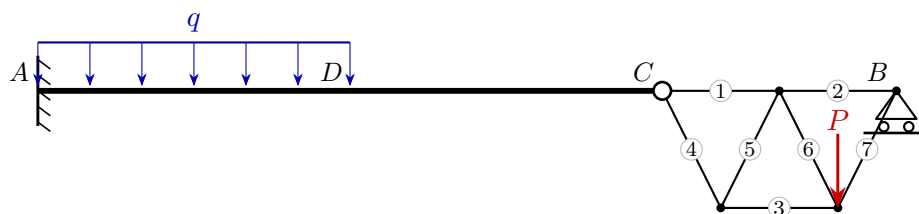
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VU Mechanik – LAWI 100515 (PI)

Exam Group A

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

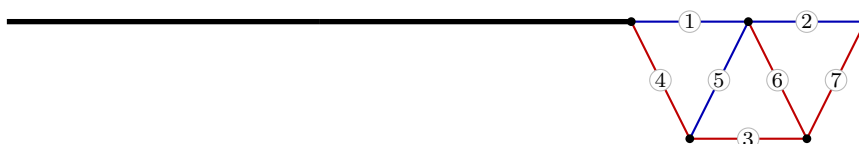
2. Support reactions and hinge forces

 $A: R_x = 0, R_y = 18.5, M = 52 \quad B: R_x = 0, R_y = 7.5$
 $C: H_x = 0, H_y = -2.5$

3. Truss member forces

Member	Force S [kN]	Type
1	-1.25	compression
2	-3.75	compression
3	2.5	tension
4	2.8	tension
5	-2.8	compression
6	2.8	tension
7	8.39	tension

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

